



DATE

22 September 2026

TO

**The Manager
The Alternative Bank
Port Harcourt, Rivers State.**

SOLAR SYSTEM AUDIT REPORT

SYSTEM

6 kW Hybrid Inverter / 10 kWh Lithium Battery

PRIMARY FINDING

Undersized Solar Panel Array

CONNECTED LOADS

- 5 Ceiling Fans
- 2 Freezers
- 1 HP Inverter Air Conditioner

OPERATING HOURS

The system is used mainly during the daytime and evening hours. The connected appliances are not operated simultaneously, and the actual load varies depending on the appliances in use.

OBSERVATION

During system inspection, it was observed that the installed solar panel capacity is inadequate for the 10 kWh lithium battery and 6 kW hybrid inverter.

The available PV generation is insufficient to provide effective battery charging during the available sunlight hours. This is causing extended charging time and reduced battery availability.

RECOMMENDATION

The existing solar panel array should be upgraded by increasing the total PV capacity to a suitable level for the inverter and battery system. This will improve charging performance and ensure the battery can be adequately charged during normal solar hours.

CONCLUSION

The low PV capacity is the main cause of the poor battery charging performance. PV array upgrade is recommended.